

# NeuroPulse AI: Electromyography (EMG) Profile

Yellow = cleaned signal | Green = muscle strength | Red = muscle active shadow

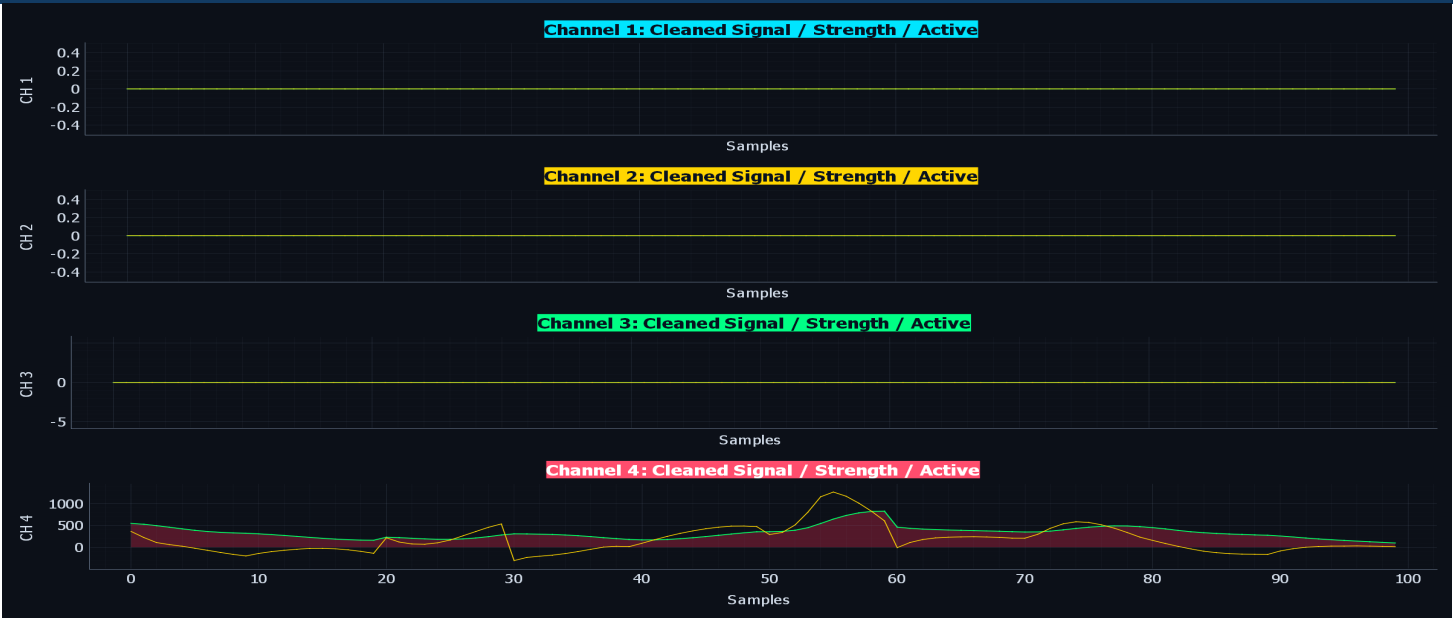
**Patient:** Patient Name | **ID:** NP-20260522-6079 | **Date:** 22-05-2026 22:10

|                     |                     |                   |                       |
|---------------------|---------------------|-------------------|-----------------------|
| Avg Strength<br>3.8 | Active Time<br>2.5% | Major Spikes<br>1 | Signal Quality<br>Low |
|---------------------|---------------------|-------------------|-----------------------|

## 1. Clinical Summary

|                         |                     |
|-------------------------|---------------------|
| Session Score           | 27/100              |
| Symmetry Score          | 0.0%                |
| Side Deficit            | 100.0% (Left lower) |
| Lowest Activity Channel | Channel 1           |
| Fatigue Drop            | Low (0.0% decline)  |
| Signal Quality          | Low                 |

## 2. 30-Second Graph Snapshot



## 3. Muscle-wise Quantitative Summary

| Muscle / Channel | Avg  | Peak  | Active % | Quality |
|------------------|------|-------|----------|---------|
| Channel 1        | 0.0  | 0.0   | 0.0%     | Low     |
| Channel 2        | 0.0  | 0.0   | 0.0%     | Low     |
| Channel 3        | 0.0  | 13.3  | 0.0%     | Low     |
| Channel 4        | 15.2 | 827.3 | 9.8%     | Low     |

## 4. Local Data-Based EMG Findings

### Signal Reasoning (Quantitative Text Analysis):

- \* Session Score: 28/100. Signal Quality: Low.
- \* Side Deficit: 100.0%. Symmetry Score: 0.0%. Marked left-side deficit detected.
- \* Lowest Activity: Channel 1. Highest Peak: Channel 4.
- \* Fatigue Drop: Low (0.0% decline from early to recent activity).
- \* Channel Balance: left-side channel balance 0%; right-side channel balance 9%.
- \* Suggested focus: compare the two right-side muscles/channels and train the weaker activation pattern.
- \* Selected affected side: Select Affected Side. Non-diagnostic rehab monitoring support only.

### Recorded Session Findings:

- \* Recording summary: 6320 EMG samples captured over approximately 60 seconds.
- \* Session Score: 27/100. Signal Quality: Low.
- \* Side Deficit: 100.0%. Symmetry Score: 0.0%. Impression: marked left-side activation deficit. Bilateral EMG symmetry is reduced and requires targeted rehab attention.
- \* Lowest Activity: Channel 1 showed the lowest average activation (0.0).
- \* Highest Peak: Channel 4 showed the highest peak activation (827.3).
- \* Channel Balance: left-side channel balance 0.0%; right-side channel balance 0.2%.
- \* Fatigue Drop: Low fatigue pattern with 0.0% decline from early to recent activity.
- \* Sustained activation: Channel 1 0.0%, Channel 2 0.0%, Channel 3 0.0%, Channel 4 9.8% above threshold.
- \* Clinical note: Affected side was not selected; interpretation is limited to channel activity and symmetry. This is a non-diagnostic rehab monitoring prototype that provides quantitative EMG feedback for physiotherapy support.

## 5. Clinical Observations & Notes

**Summary Notes:** N/A

### Comparative Analysis:

- Left Side Avg: 0.0
- Right Side Avg: 7.6
- Symmetry Ratio (L/R): 0.0%
- Most Dynamic Channel: Channel 4